

Pancreatic Cystic Lesions and Malignancy: Assessment, Guidelines, and the Field Defect

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Abbreviations: IPMN = intraductal papillary mucinous neoplasm, MCN = mucinous neoplasm, MPD = main pancreatic duct, MRCP = MR cholangiopancreatography, PNET = pancreatic neuroendocrine tumor, SCA = serous cystadenoma, SPN = solid pseudopapillary neoplasm

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Recognize common cystic pancreatic lesions.
- List the features of pancreatic cystic lesions that are more commonly associated with malignancy.
- Understand the concept of field defect related to pancreatic cysts.

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The widespread use of high-spatial-resolution cross-sectional imaging has led to an increase in detection of incidental pancreatic cystic lesions. These lesions are a diverse group, ranging from indolent and premalignant lesions to invasive cancers. The diagnosis of several of these lesions can be suggested on the basis of their imaging appearance, while many other lesions require follow-up imaging and/or aspiration. The smaller cystic lesions, often branch-duct intraductal papillary mucinous neoplasms, have overlapping imaging characteristics that make diagnostic assessment of the natural history and malignancy risk confusing. Expert panels have developed societal guidelines, based on a consensus, for surveillance of these lesions. However, these guidelines are often inconsistent and are constantly evolving as additional scientific data are accumulated. Identification of features associated with increased risk of malignancy is important for proper management. The concept of field defect, whereby pancreatic adenocarcinoma develops at a site different from the site of the pancreatic cyst, adds to the complexity of screening guidelines. As a result of the differences in guidelines, key stakeholders (eg, radiologists, gastroenterologists, and surgeons) must review and come to a consensus regarding which guideline, or combination of guidelines, to follow at their individual institutions. Standardized reporting and macros are helpful for ensuring the uniformity of interpretations. Radiologists play a critical role in the detection and characterization of pancreatic cystic lesions, in the follow-up recommendations for these lesions, and in the detection of associated cancer.

An invited commentary by Zaheer is available online.

Online supplemental material is available for this article.

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Introduction

The widespread use of high-spatial-resolution cross-sectional imaging has led to increased rates of detecting pancreatic cystic lesions. The spectrum of these lesions ranges from benign entities (pseudocyst, inflammatory cyst) to potentially malignant neoplasms (mucinous neoplasm [MCN], intraductal papillary mucinous neoplasm [IPMN]) and frankly malignant tumors (solid pseudopapillary neoplasm [SPN], pancreatic neuroendocrine tumor [PNET], and ductal adenocarcinoma) (1). Most cystic lesions of the pancreas are benign and incidentally detected with imaging in approximately 10% of the population and in up to 30% of individuals older than 70 years (2).

TEACHING POINTS

- The imaging features of IPMNs that were reported as “worrisome” for malignancy in the 2017 international consensus Fukuoka guidelines and suggest the need for further investigation with endoscopic US include a cyst diameter greater than 3 cm, an enhancing mural nodule smaller than 5 mm, a thickened enhancing cyst wall, MPD dilatation of 5–9 mm, an abrupt change in MPD diameter with upstream pancreatic atrophy, lymphadenopathy, and a cyst growth rate of greater than or equal to 5 mm per 2 years.
- High-risk stigmata considered by the International Association of Pancreatology to be indications for resection include obstructive jaundice, a 5-mm or larger enhancing mural nodule, and MPD dilatation of 1 cm or greater without another cause of obstruction.
- Assessing the risk of a single pancreatic cyst developing into invasive cancer can be challenging, and the presence of an IPMN may portend an increased risk of invasive pancreatic cancer at disparate sites within the pancreas.
- Multiple societies have created guidelines for the management of pancreatic cysts. Five major guidelines are the international consensus Fukuoka 2017, American College of Gastroenterology 2018, American College of Radiology 2017, European Study Group 2018, and American Gastroenterology Association 2015 guidelines. However, as shown in Table E1, there is variability in the guidelines among the different societies.
- As a result of the differences in guidelines, key stakeholders (eg, radiologists, gastroenterologists, and surgeons) must review and come to a consensus regarding which guideline (or combination of guidelines) to follow at their individual institutions.

The assessment of pancreatic cystic lesions should include careful assessment of imaging features (cyst morphology, presence of duct communication) and clinical findings including cystic fluid analysis, when available, to improve diagnostic accuracy. Identification of certain key imaging findings that confidently suggest one lesion rather than the other, and more important, of features that are more often associated with malignancy, is crucial, as these features are used to determine proper management. The biologic behavior and risk of malignant transformation in select pancreatic cysts vary according to the histologic subtype. A relevant possibility associated with pancreatic cystic lesions, specifically IPMNs, is the risk of a field defect—that is, the development of a pancreatic adenocarcinoma at a location within the pancreas that is different from the site of the cystic lesion. The adenocarcinoma can occur concomitantly with the identification of the cystic lesion (synchronously) or later (metachronously). Many societies have developed expert but nonuniform consensus clinical guidelines, which are continuously evolving according to accumulating medical evidence, to facilitate decisions regarding management and surveillance of these lesions.

In this article, we describe an approach to assessment of some of the predominant pancreatic cystic lesions, with an emphasis on features suggestive of malignancy. Benign inflammatory cysts such as pseudocysts and lymphatic cysts are not discussed, as they are beyond the scope of this article. However, in the setting of pancreatitis, a pseudocyst or walled-off necrosis should be considered a cause for cystic masses.

We also review the current guidelines for surveillance of pancreatic cysts, highlighting the complexity of and controversies regarding managing and following up these lesions. Finally, we discuss the importance of the field defect related to development of invasive pancreatic adenocarcinomas.

Background and Epidemiologic Factors

Pancreatic cystic lesions are commonly detected incidentally because of the increased use of and improvements in cross-sectional imaging, as well as an aging population. They are reported on 20%–44% of MRI images of the abdomen (3,4) and on up to 3% of CT scans (5). The higher rate of detection with MRI is due to the high contrast resolution of T2-weighted MR images, which better delineate even very small cystic lesions.

In a meta-analysis of 48 860 asymptomatic patients undergoing abdominal imaging, a prevalence of 8% of patients were noted to have pancreatic cysts (6). A large population-based study from Northern Germany showed a weighted prevalence of pancreatic cysts of 49.1% at whole-body MRI (7). In high-risk populations, including patients with a family history of pancreatic cancer, pancreatic cysts have been reported in up to one-third of individuals (8). It has been reported that individuals with one, two, or three or more first-degree relatives with pancreatic cancer have a 4.6, 6.4, and 32.0-fold increased risk of pancreatic cancer, respectively. As such, *familial pancreatic cancer* has been defined as pancreatic cancer affecting at least two first-degree relatives (9).

Imaging Features and Practical Approach

The frequent detection of incidental pancreatic cysts necessitates an organized systematic approach to evaluating these lesions with imaging. The key features of pancreatic cysts that should be evaluated include size, number of cysts, presence or absence of septa, enhancing cyst walls and nodules, and relationship to the main pancreatic duct (MPD). These findings may guide management, including follow-up imaging, referral to a specialist for endoscopic US and aspiration, and/or surgical removal.

Communication with the MPD is a critical distinguishing feature of pancreatic cysts (10). Similarly, the presence of a ductal communication may suggest a pseudocyst or walled-off necrosis in the setting of pancreatitis rather than other lesions such as MCN. Many small cystic lesions do not show a communication and are usually presumed to be branch-duct IPMNs and undergo a follow-up similar to that for these mucinous neoplasms.

It is important to evaluate the associated features of the cyst. An internal enhancing solid component increases the suspicion for malignancy. A thick enhancing cyst wall is another feature that can suggest a malignant or potentially malignant lesion such as PNET or adenocarcinoma (which can rarely appear cystic) or MCN, instead of the typical thin-walled cyst in branch-duct IPMN. MRI is often useful in evaluating internal complexity, as it enables better characterization of the cyst content, including the presence of hemorrhage, enhancing nodules, and septa. Usually, diffusion-weighted imaging is not helpful for differentiating neoplastic from nonneoplastic cystic lesions, and its utility in determining potentially malignant lesions has not been established (11).

Demographic data can be useful, but they should not be the predominant information used to characterize pancreatic cystic lesions (Table) (8,12–15). A common mnemonic based on patient age that can help guide the differential diagnosis is that SPN is a “daughter” lesion; MCN, a “mother” lesion; and serous cystadenoma (SCA), a “grandmother” lesion (12). Although this mnemonic serves as a broad guideline, it is certainly not definitive, given the significant overlap in features. Demographic data are most specific in the setting of MCNs, as these lesions are rare in males. Gender is usually not associated with an increase in the malignant potential of pancreatic cystic lesions, with the apparent exception of MCNs, whereas advanced age has been correlated with increased malignant risk (16,17).

Pancreatic Cystic Lesions

The typical clinical and imaging characteristics of the described pancreatic cystic lesions, as well as the classic biochemical and molecular findings at cyst aspirate fluid analysis, are summarized in the Table (8,12–15). The main pancreatic cystic lesions discussed in this article are depicted in Figure 1.

Serous Cystadenoma

SCAs are benign lesions that can mimic other benign and malignant lesions. However, they

potentially can be diagnosed noninvasively when they demonstrate a characteristic imaging appearance, especially on MR images. While SCAs typically occur sporadically, there is an increased incidence in patients with von Hippel–Lindau syndrome. Most affected patients (up to 80%) are asymptomatic at the time of diagnosis, with lesions found incidentally (18).

Larger SCAs can cause symptoms (18). The microcystic form has classic imaging features, with multiple (>6) small (<2 cm) cysts that result in a multilobulated external contour that is not generally seen in other cystic lesions (Fig 2). The honeycomb pattern with numerous tiny cysts may appear solid on contrast-enhanced CT or MR images owing to enhancement of the closely apposed small cyst walls and can mimic a hypervascular neuroendocrine tumor. Similarly, at US, these tiny cysts may have a solid appearance as a result of the numerous interfaces from the small cyst walls. SCA and cystic PNET are among the few hypervascular cystic pancreatic neoplasms (19).

T2-weighted MRI can be helpful in showing the cystic components of cystic pancreatic neoplasms, especially the microcystic form. It is not uncommon for large SCA lesions (15%–20%) to consist of a combination of small and large cysts (20). An enhancing scar with stellate calcifications is seen at CT in up to 30% of lesions and is considered pathognomonic for SCA (12).

SCAs can be diagnosed with high specificity. Challenges to diagnosis occur when an SCA appears to be solid or has calcifications, mimicking PNET; when oligocystic SCAs (lesions with fewer cysts) (Fig 3) mimic the more malignant MCN; or rarely, when SCA with intratumoral hemorrhage mimics PNET or SPN. SCAs may have a morphologic appearance that is similar to that of branch-duct IPMNs, but they do not communicate with the pancreatic duct. This distinction is significant, considering the association of branch-duct IPMNs with cancer. However, not all branch-duct IPMNs show a communication with the pancreatic duct. An additional challenge is that although SCAs are benign, they may cause mass effect and upstream pancreatic duct dilatation. Generally, asymptomatic SCAs do not require follow-up imaging unless the diagnosis is uncertain. Surgery is considered when patients experience symptoms related to the compression of other organs (13).

Mucinous Cystic Neoplasms

MCNs are seen nearly exclusively in women and should be suspected especially in middle-aged women who have typical imaging features, especially in the absence of a history of pancreatitis.

Typical Demographic, Morphologic, and Biochemical or Molecular Features of Pancreatic Cystic Lesions

Feature	SCA	MCN	SPN	Cystic PNET	IPMN
Gender	More common in females than in males	Almost exclusively in females	More common in females than in males	Equally common in males and females	Equally common in males and females
Age (decade)	Variable, usually 5th–7th decade (“grandmother” lesion)	Variable, usually 4th–5th decade (“mother” lesion)	2nd–4th decade* (“daughter” lesion)	5th–6th decade	Variable, usually 6th–7th decade
Location	More common in head than in body and tail	Body and tail	Body and tail	Body and tail	BD: all regions
Morphology	Classic microcystic or “honeycomb” appearance, lobulated	Few large cysts, cyst within cyst, thick wall with or without septa	Cystic and/or solid* (necrosis or hemorrhage), oval, large*	Cystic and solid (degeneration and/or necrosis)	BD: dilated side branch MD: segmental or diffuse dilatation of PD
Cyst size	<2 cm	>2 cm	Variable	Variable	Variable
Number of loculations or cysts	>6	<6	Unilocular, thick capsule	Unilocular, thick enhancing wall*	BD: variable
CT	Multicystic, lobulated with enhancing septa	Few cysts with occasional mural nodule and/or septa	Heterogeneous solid and/or cystic with hyperattenuating areas from hemorrhage	Cystic with smooth margins, hypervascular peripheral rim	BD: cysts communicate with PD MD: segmental or diffuse dilatation of PD
MRI	T1W: low SI T2W: high SI Central scar	T1W: usually low SI T2W: high SI; may have thick enhancing walls, septa, and/or nodules	T1W and T2W: variable owing to internal necrosis and hemorrhage*	T1W: low SI T2W: high SI Hypervascular peripheral rim	T1W: low SI T2W: high SI Enhancing nodules increase likelihood of malignancy
Calcification	30%–40% involve central scar	Up to 25%, wall or septal	Up to 30%, predominantly peripheral and punctate	May be present	Can be seen, likely related to mucin
MPD communication	No	No	No	No	Yes
Molecular analysis	<i>VHL</i> positive	<i>KRAS</i> , <i>RNF3</i> positive (<i>GNAS</i> negative)	<i>CTNNB1</i> positive	<i>MEN1</i> , <i>DAXX</i> , <i>ATRX</i> positive	<i>KRAS</i> , <i>GNAS</i> , <i>RNF43</i> positive
Fluid analysis	Low CEA, low amylase, high VEGF	Mucinous, high CEA,* variable amylase	Low CEA, low amylase	Low CEA, low amylase	Mucinous, high CEA,* high amylase

Sources.—References 8, 12–15.

Note.—BD = branch duct, CEA = carcinoembryonic antigen, MD = main duct, PD = pancreatic duct, SCA = serous cystadenoma, SI = signal intensity, T1W = T1-weighted MRI, T2W = T2-weighted MRI, VEGF = vascular endothelial growth factor.

*Feature is classically associated with the given tumor type.

MCNs are usually (95% of cases) located in the body or tail of the pancreas (21). The histologic diagnosis is based on the presence of ovarian stromata (22). These lesions are extremely rare in male patients, with approximately 12 cases of MCNs with ovarian-like stromata reported (10).

It is important to distinguish MCNs from other lesions, including pseudocysts, as they are considered to require surgery on the basis of their malignant potential. Unlike SCAs, MCNs tend to have fewer (<6) and larger (>2 cm) cysts (Fig

4), and they lack external lobulations. These neoplasms may have a “cysts-within-a-cyst” appearance. Like IPMNs, MCNs are found to contain mucin at aspiration, but unlike IPMNs, they do not communicate with the MPD. They often have septa or nodules on imaging studies and may have capsular calcifications that are best appreciated at CT.

The malignancy rates associated with MCNs range from 10% to 17% (23). Features associated with malignancy include invasion of adja-

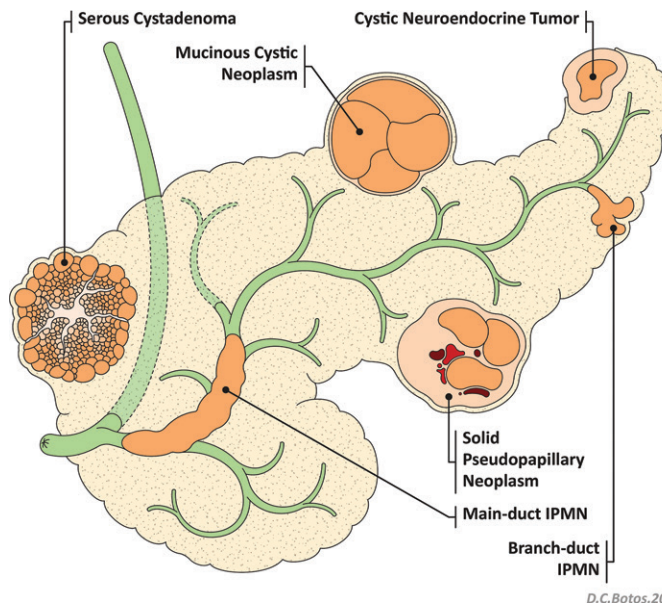


Figure 1. Diagram of pancreatic cystic lesions, including SCA, MCN, IPMN, SPN, and cystic PNET.

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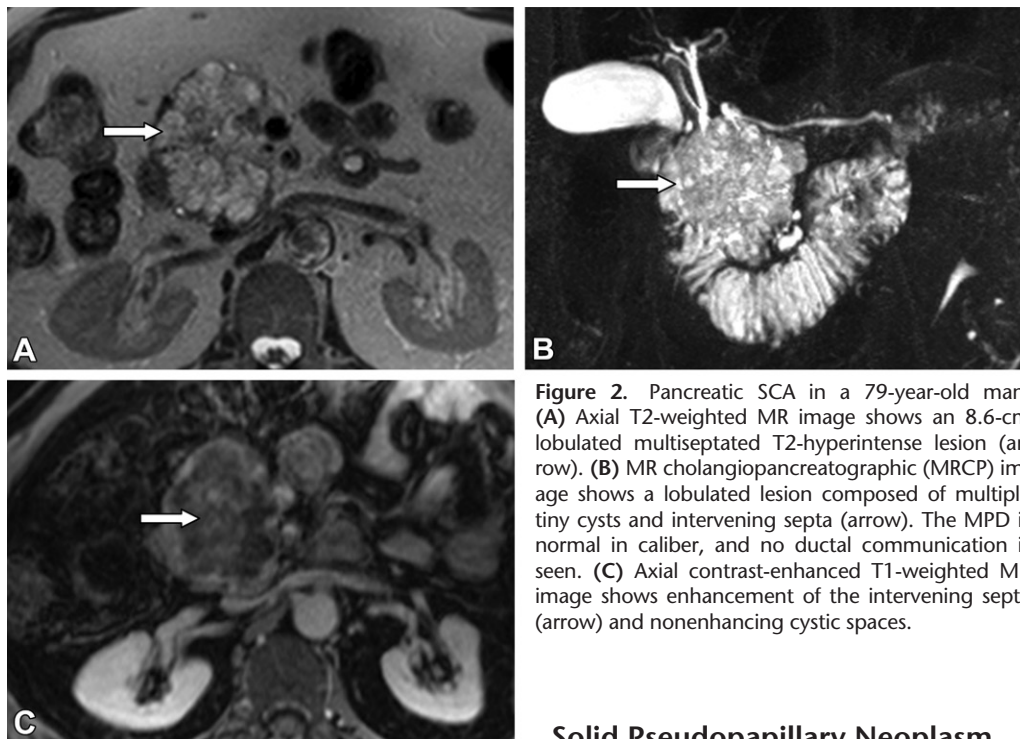


Figure 2. Pancreatic SCA in a 79-year-old man. (A) Axial T2-weighted MR image shows an 8.6-cm lobulated multiseptated T2-hyperintense lesion (arrow). (B) MR cholangiopancreatographic (MRCP) image shows a lobulated lesion composed of multiple tiny cysts and intervening septa (arrow). The MPD is normal in caliber, and no ductal communication is seen. (C) Axial contrast-enhanced T1-weighted MR image shows enhancement of the intervening septa (arrow) and nonenhancing cystic spaces.

cent structures, nodal and distant metastases, larger lesion size (cyst ≥ 4 cm), thick irregular walls, enhancing soft-tissue components or mural nodules, peripheral calcifications, and pancreatic duct dilatation (Fig 5) (24–26). An MCN in a male should be more concerning for malignancy (17). Metastases, when present, occur mainly in the liver and are typically cystic (Fig 6) (24). Among MCNs smaller than 4 cm and without worrisome features, malignant transformation has occurred in 0.03% of cases (24).

Solid Pseudopapillary Neoplasm

SPN, also referred to as solid pseudopapillary epithelial neoplasm, is a rare pancreatic tumor that occurs predominantly in young women and accounts for 1%–2% of all pancreatic exocrine tumors. They are considered low-grade malignant neoplasms and are removed surgically because of the risk of malignancy. SPNs are associated with a good prognosis, with a 5-year survival rate of 71.1%–97.0% (27).

The complex cystic components of SPNs comprise hemorrhagic and necrotic material, and their imaging features vary on the basis of these components (Fig 7). SPNs tend to be large (2.5–17.0 cm; mean size, 9 cm) at presentation

Figure 3. Pancreatic oligocystic SCA in a 70-year-old woman. Axial T2-weighted MR image shows a 10-cm unilocular hyperintense lesion with a lobulated contour (arrow) and internal septa, mimicking MCN.

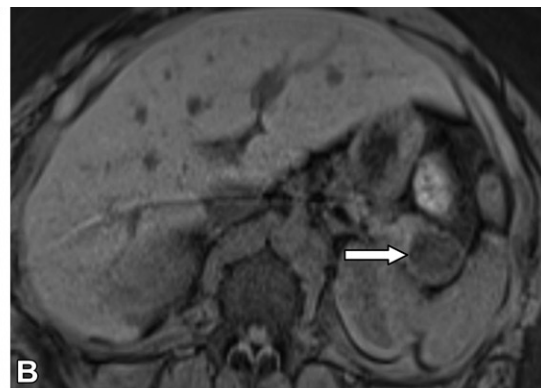
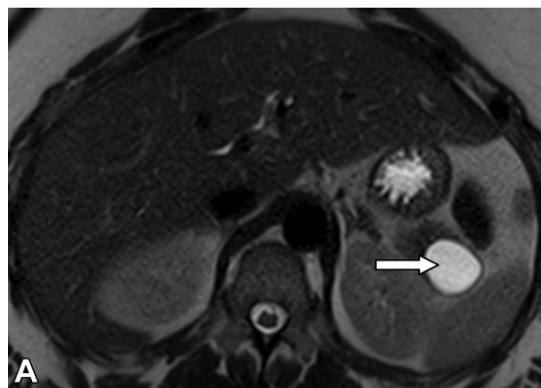
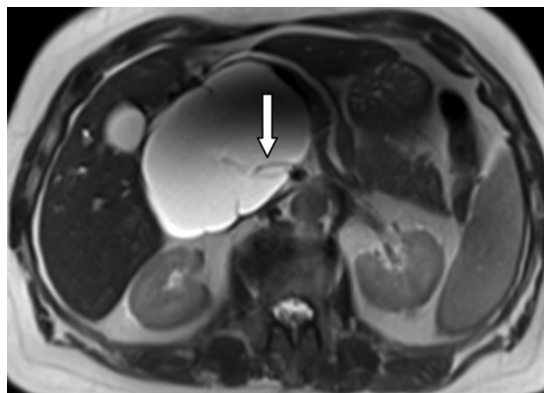
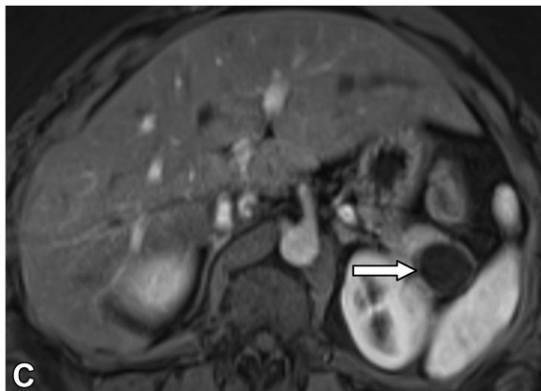


Figure 4. MCN with low-grade dysplasia in a 43-year-old woman. Axial T2-weighted (A), noncontrast T1-weighted (B), and contrast-enhanced T1-weighted (C) MR images show a 2.2-cm well-circumscribed unilocular cystic lesion (arrow) in the pancreatic tail. The lesion is hyperintense on the T2-weighted image (A), is hypointense on the noncontrast T1-weighted image (B), and does not enhance on the contrast-enhanced T1-weighted image (C).



and well defined, can be found anywhere in the pancreas, and may have calcifications. MRI is helpful in confirming the presence of internal hemorrhage, narrowing the differential diagnosis, and increasing suspicion for SPN, especially in the appropriate demographic group of young women (28,29). Hemorrhage in SPN occurs owing to rupture of the lesion's inadequately supported vasculature (28) and appears as high signal intensity on T1-weighted MR images. Hemorrhage is seldom seen in other lesions such as SCAs, walled-off necrosis, and bleeding from underlying pseudoaneurysms. Tumor recurrences or metastases after surgery have been reported in 4.4%–9.0% of patients (30). The most common location of metastases is the liver (Fig 8). Rarely, SPN metastases occur in the lymph nodes, peritoneum, or lungs (27,29). Malignant SPNs more likely have the following features: size larger than 5–6 cm, location in the tail of the pancreas, focal discontinuity of the

capsule, and liver lesions suggestive of metastases (29).

Cystic PNETs

PNETs are uncommon pancreatic malignancies that typically occur sporadically but also can be associated with multiple endocrine neoplasia type 1, neurofibromatosis, or von Hippel-Lindau syndrome. PNETs are divided into two groups: (a) syndromic or hyperfunctioning/functional; and (b) nonsyndromic or nonfunctional (31). Functional PNETs are usually subcentimeter lesions and are brought to clinical attention by symptoms caused by the hormones produced by these tumors. On the other hand, nonfunctional tumors usually manifest when

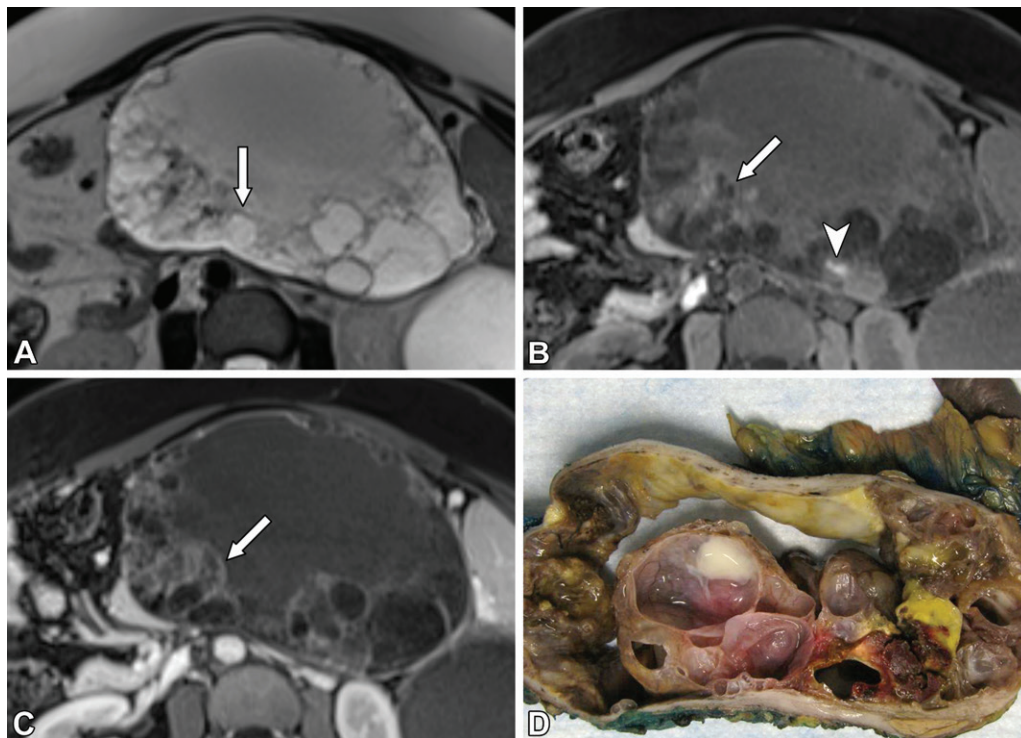


Figure 5. MCN in a 34-year-old woman. (A) Axial T2-weighted MR image shows a complex T2-hyperintense cystic lesion composed of multiple smaller T2-hyperintense cystic locules (arrow). (B) Axial noncontrast T1-weighted MR image shows locules (arrow) and hyperintense foci (arrowhead) consistent with areas of hemorrhage. (C) Axial contrast-enhanced T1-weighted MR image shows enhancing septa and nodules (arrow). (D) Gross specimen of the resected malignant MCN shows a multiloculated cystic mass.

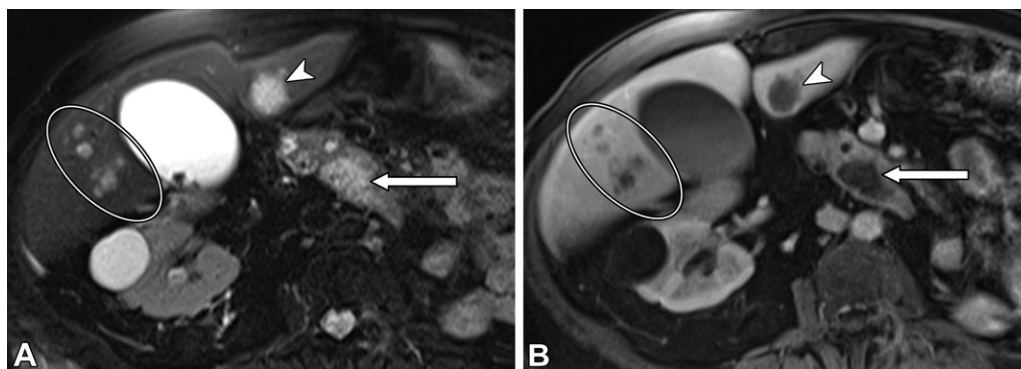


Figure 6. Mucinous adenocarcinoma in a 72-year-old woman with a carcinoembryonic antigen level of 104.1 ng/mL and a history of breast cancer. (A) Axial T2-weighted MR image shows a 3.2-cm lobulated T2-hyperintense lesion (arrow) within the uncinate process and T2-hyperintense liver metastases (arrowhead and oval outline). (B) Axial contrast-enhanced T1-weighted MR image shows a hypoenhancing lesion (arrow) and multiple hypoenhancing liver metastases (arrowhead and oval outline). Biopsy yielded mucinous adenocarcinoma, accounting for the hyperintensity seen at T2-weighted MR imaging.

they are several centimeters in size, having remained clinically occult until they became large enough to cause mass effect (31). Cystic PNETs account for 7%–17% of all resected PNETs (32). Owing to their malignant potential, it is important to distinguish these tumors from other pancreatic cystic lesions.

The term *cystic PNET* was originally attributed to large tumors arising from a solid lesion that had undergone cystic degeneration owing to restric-

tion of internal blood flow or internal hemorrhage. In this original categorization of cystic PNETs, lesions tended to be large (8.4 vs 2.9 cm) (33) and mostly nonfunctioning (91% of cases) (34) compared with their solid counterparts. However, more recent literature describes a distinct cystic subtype of PNETs that has been detected increasingly over the past decade owing to increased use of high-spatial-resolution cross-sectional imaging (32). These smaller cystic subtypes of PNET may

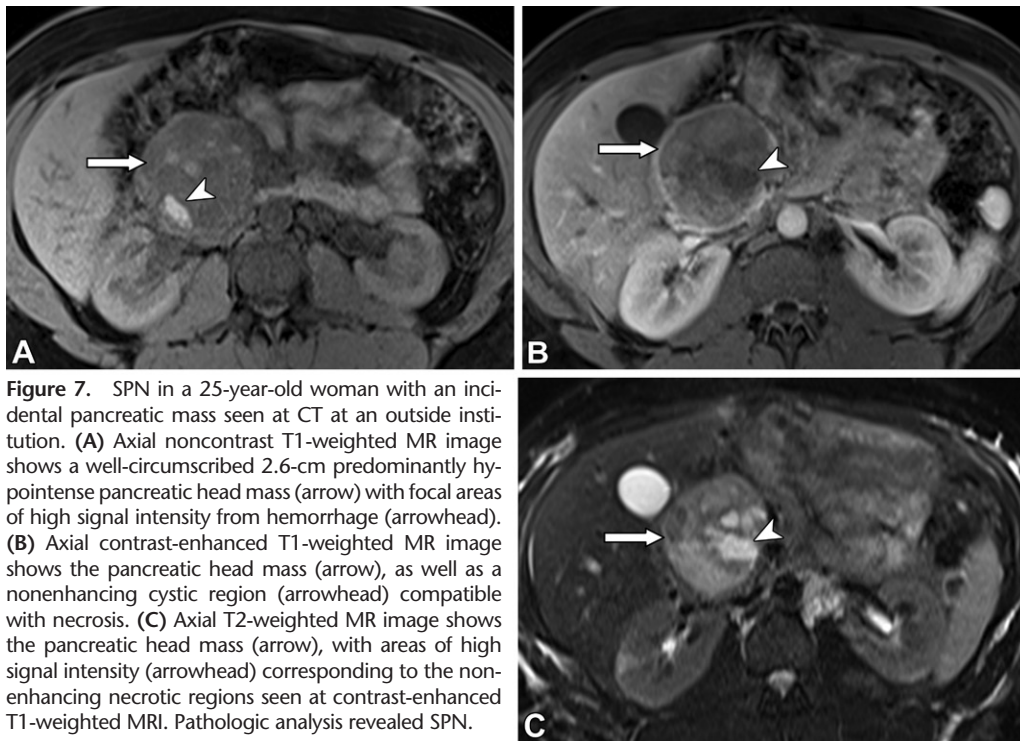


Figure 7. SPN in a 25-year-old woman with an incidental pancreatic mass seen at CT at an outside institution. (A) Axial noncontrast T1-weighted MR image shows a well-circumscribed 2.6-cm predominantly hypointense pancreatic head mass (arrow) with focal areas of high signal intensity from hemorrhage (arrowhead). (B) Axial contrast-enhanced T1-weighted MR image shows the pancreatic head mass (arrow), as well as a nonenhancing cystic region (arrowhead) compatible with necrosis. (C) Axial T2-weighted MR image shows the pancreatic head mass (arrow), with areas of high signal intensity (arrowhead) corresponding to the non-enhancing necrotic regions seen at contrast-enhanced T1-weighted MRI. Pathologic analysis revealed SPN.

represent their own distinct entity. It has been proposed that these subtypes possess their own distinct clinicopathologic features that distinguish them from other PNETs. Specifically, this subtype occurs more commonly in the pancreatic body and tail, is typically nonfunctioning, demonstrates less aggressive behavior (lower ki-67 index), and is associated with a lower rate of metastases (32).

The imaging features of cystic PNETs overlap with those of other cystic pancreatic lesions. However, owing to their abundant blood supply, they should be suspected when there is an avidly enhancing wall (Fig 9). The hypervascular peripheral wall is considered the most suggestive radiologic characteristic of cystic PNETs and is better appreciated with use of a dedicated pancreas protocol CT (33) or MRI.

With MRI, the internal cystic component demonstrates low signal intensity on T1-weighted images and high signal intensity on T2-weighted images. Somatostatin-receptor PET/CT has been shown to be an excellent imaging modality to locate the tumor, assess for metastases, and affect management by identifying additional sites of disease when surgery is contemplated (35). Gallium 68–tetraazacyclododecane tetraacetic acid-octreotate PET/CT can help detect PNET (with 93% sensitivity and 91% specificity), stage PNET, detect recurrences, and assess tumor heterogeneity, especially that of grade 2 and grade 3 tumors (36). Cystic PNETs have been reported to demonstrate less aggressive behavior compared with solid PNETs. They have also been shown

to be less likely to involve regional lymph node metastases, vascular invasion (34,37), perineural invasion and synchronous distant metastasis (34,38), or recurrence during follow-up (38).

Intraductal Papillary Mucinous Neoplasm

IPMNs are characterized by a dilated MPD and/or dilated branch ducts with papillary projections composed of dysplastic mucinous epithelium (39). Many cases of IPMN are detected incidentally and are asymptomatic, with prevalences ranging from 0.0008% to 10.0% in patients older than 70 years (40). According to the latest World Health Organization recommendations, dysplastic grades of IPMNs should be identified as low-, intermediate-, or high-grade dysplasia (replacing the term *carcinoma in situ*) (41), and invasive carcinoma (42). Histologically, IPMNs can be further classified as intestinal, pancreatobiliary, gastric, and oncocytic subtypes (43). On the basis of the type of ductal involvement, IPMNs can be categorized as branch-duct, main-duct, or mixed-type IPMNs.

Branch-Duct IPMN.—Branch-duct IPMNs account for 41%–64% of IPMNs (44) and appear as a cluster of small cysts with lobulated margins and septa (grapelike lobulated appearance), or unilocular cysts on imaging studies. On the basis of its frequent location in the uncinate process, IPMN should be suspected when a cystic lesion in the uncinate process is encountered. Unlike SCA, branch-duct IPMN communicates with the pancreatic duct (Fig 10). This communication

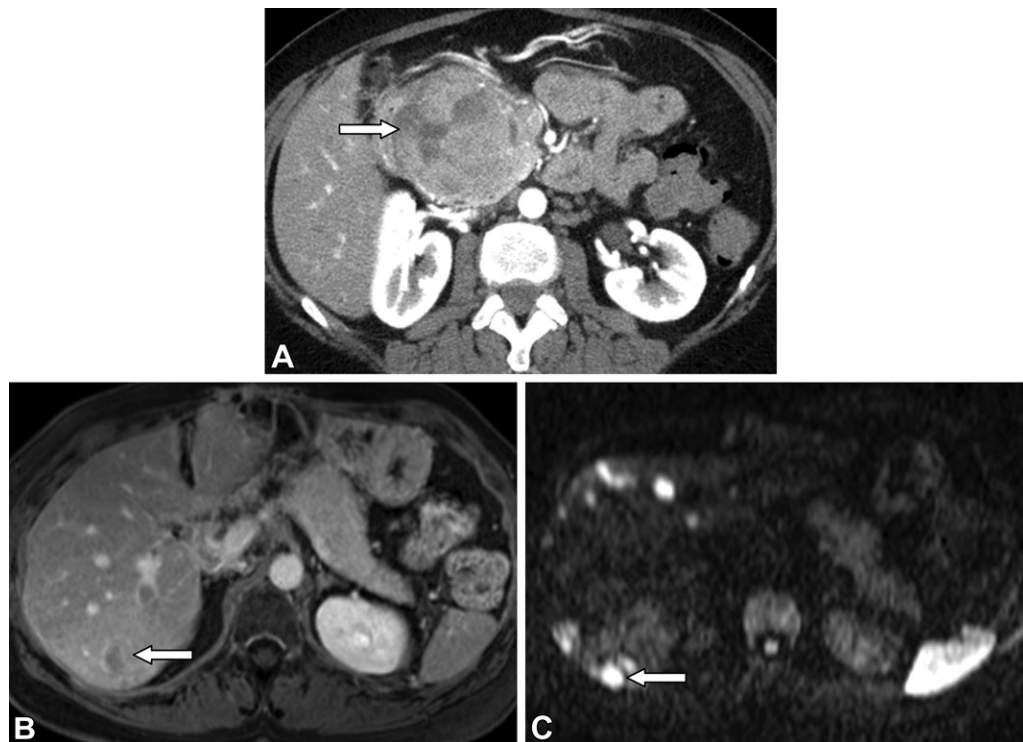


Figure 8. SPN with perineural invasion seen at surgical-pathologic analysis in a 54-year-old woman. (A) Axial contrast-enhanced CT image shows a heterogeneously enhancing pancreatic head mass (arrow). (B) Axial contrast-enhanced T1-weighted MR image 2 years after surgery shows hypoenhancing liver lesions (arrow) representing metastases, which are best seen on the axial diffusion-weighted MR image, demonstrating restricted diffusion (arrow in C). (C) Multiple additional liver metastases are best seen on the diffusion-weighted image. (Case courtesy of Kiran Thakrar, MD, North Shore University Health System, Chicago, Ill.)

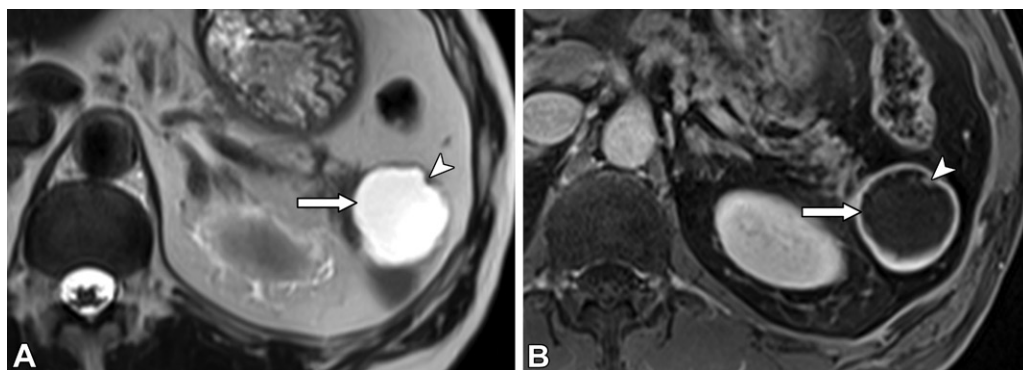


Figure 9. Cystic PNET in a 66-year-old man with a history of prostate cancer, and a pancreatic cyst seen at CT. (A) Axial T2-weighted MR image shows a 4.2-cm cystic lesion (arrow) in the pancreatic tail, with subtle nodularity (arrowhead) along the cyst wall. (B) Axial contrast-enhanced T1-weighted MR image shows a thick enhancing wall and nodularity (arrowhead) of the cystic lesion (arrow). Biopsy yielded World Health Organization grade 1 (carcinoid) neuroendocrine tumor.

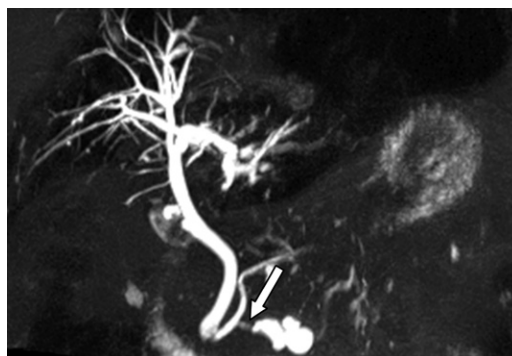


Figure 10. Branch-duct IPMN in a 77-year-old man. MRCP image shows a 3.2-cm lesion (arrow) communicating with the main duct, compatible with branch-duct IPMN.

can be seen on CT images but is more commonly and better visualized on MR images. Accurately distinguishing small branch-duct IPMNs from MCN and SCA (particularly the oligocystic subtype) on imaging studies can be challenging if

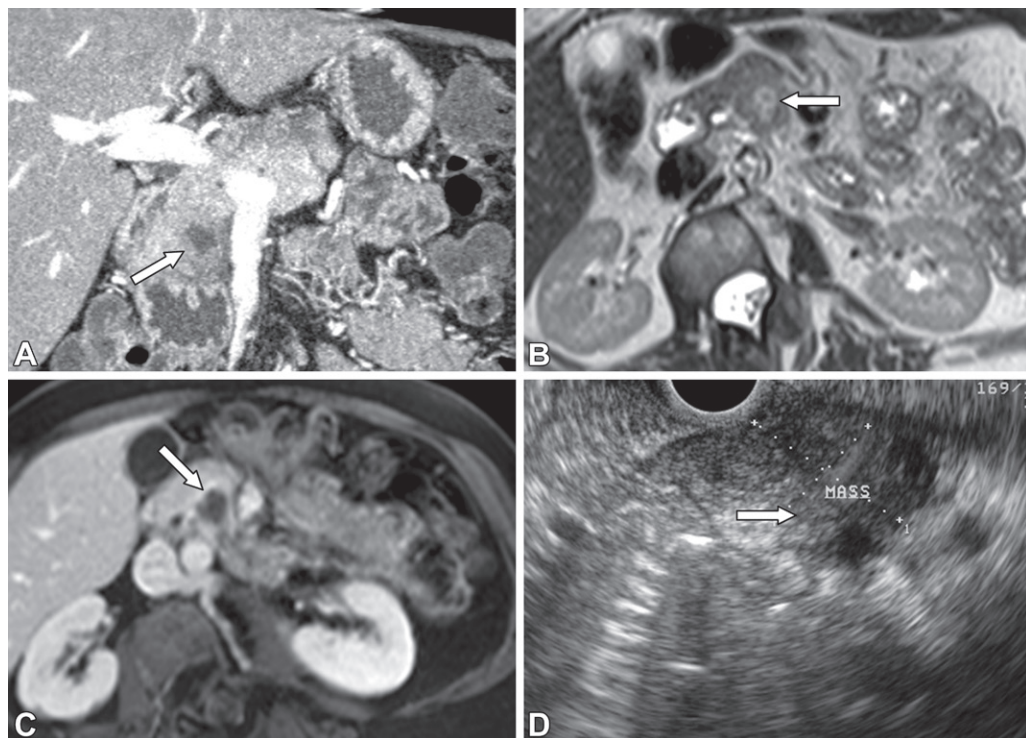


Figure 11. Pancreatic adenocarcinoma in a 56-year-old woman with a hypoattenuating mass in the uncinate process of the pancreas mimicking branch-duct IPMN at CT. (A) Coronal CT image shows a 2.2-cm hypoattenuating cyst-appearing mass (arrow) in the pancreatic head, which could have been easily confused with a branch-duct IPMN if it was not evaluated closely, but the radiologist reading the scan believed it to be suspicious. (B) T2-weighted MR image obtained for further evaluation shows a mildly hyperintense lesion (arrow) that was not characteristic of a cyst. (C) Axial contrast-enhanced T1-weighted MR image shows a cyst-appearing lesion with prominent rim enhancement (arrow), further suggesting that this did not represent a cyst. (D) Endoscopic US image shows an echogenic predominantly solid mass (arrow). The patient underwent a Whipple procedure, and surgical-pathologic analysis revealed poorly differentiated pancreatic adenocarcinoma.

the communication with the duct is not visualized. Branch-duct IPMNs are multiple in 40% of cases; thus, when present, they can suggest IPMN rather than other lesions. MRI can also be helpful, as small masses that are indeterminate or suspected to be branch-duct IPMNs at CT may be shown to be solid masses such as adenocarcinomas at MRI or endoscopic US (Fig 11).

The histologic subtypes of IPMN are associated with the different morphologic types and varying frequencies of malignancy. Gastric subtypes are more often associated with branch-duct IPMNs and have the lowest association with malignancy (45). Pancreatobiliary subtypes are associated with the highest rates of tumor invasion (68%) and transformation to tubular carcinoma (82%) (40,45). The incidence of malignancy, including both IPMN-induced carcinoma and concomitant pancreatic adenocarcinoma, associated with branch-duct IPMNs has been reported to be 3.3% at 5 years and 15.0% at 15 years (46). Tanaka (47) reported that on average, branch-duct IPMNs are associated with noninvasive cancer in 10.1% of cases and with invasive cancer in 13.1% of cases. However, these findings are based on surgical series. The actual frequencies

of invasive and noninvasive cancers are markedly lower, as the majority of small incidental branch-duct IPMNs are not resected.

Main-Duct and Mixed-Type IPMNs.—Main-duct IPMNs constitute up to 15%–21% of IPMNs (44) and are characterized by segmental or diffuse dilatation of the MPD caused by the tumor's mucin production. Main-duct IPMNs may appear to be cystic if only a short segment of the MPD is involved. A soft-tissue nodule representing the mucin-producing tumor may be difficult to identify because of its small size. Guidelines differ regarding the size of the MPD that suggests main-duct IPMN when other causes of dilatation are not seen. The international consensus Fukuoka guidelines use a lower cutoff of 5 mm to increase sensitivity for main-duct IPMN (42), while the American College of Radiology white paper suggests 7 mm as the cutoff (Fig 12) (23). Regardless of which cutoff is used to diagnose a main-duct IPMN, increasing diameter of the MPD is associated with an increased risk of malignancy (46). Using a lower cutoff may introduce challenges, as there is overlap with other causes of MPD dilatation, including ectasia from



Figure 12. Main-duct IPMN in an 88-year-old woman. MRCP image shows a diffusely dilated 1.7-cm pancreatic duct (arrow) in the setting of main-duct IPMN.

chronic pancreatitis, other causes of glandular atrophy and duct dilatation, and ductal adenocarcinoma. Therefore, it is important to exclude other etiologic factors, such as a pancreatic mass, ductal stone, or stricture, as the cause for the dilatation.

In contrast to an underlying tumor or stone, main-duct IPMN will taper when it is focal and without associated abnormalities, but when the entire duct is involved, it may not taper and further workup is often required. A patulous ampulla with bulging of the papilla into the duodenal lumen, analogous to endoscopic retrograde cholangiopancreatography findings, is uncommonly seen on cross-sectional images but is highly suggestive of diffuse involvement of main-duct IPMN (28). Small papillary projections can be helpful as confirmatory findings but typically are difficult to identify on CT or MR images and are better seen with endoscopic US and suggestive of invasive carcinoma or high-grade dysplasia (42). Pancreatic atrophy may be present and may mimic or be associated with chronic pancreatitis.

Main-duct IPMN is associated with a higher risk of malignancy than is branch-duct IPMN (47). According to Tanaka's (47) review of the literature, approximately 20% of main-duct IPMNs are associated with noninvasive cancer and 40% are associated with invasive cancer; these findings can prompt surgical consideration (47).

Recent data have shown that even among main-duct IPMNs, there may be different size cutoffs for malignancy, depending on the location of the main-duct IPMN. Crippa et al (48) found an optimal threshold for malignancy of greater than or equal to 9 mm in the head and of greater than or equal to 7 mm in the body and tail of the pancreas. Up to 40% of surgically resected IPMNs with invasive carcinoma will recur within 12 months, and the reported associated 5-year survival rate is 36%. The most common sites for recurrence are other sites in the pancreas and distant metastases (including the liver and lymph nodes) (49,50).

Mixed-type IPMN demonstrates characteristics of both main-duct IPMN and branch-duct IPMN and is observed in 22%–38% of cases (44). When evaluating an individual cystic lesion suspected of being a branch-duct IPMN, it is important to evaluate the MPD to exclude a mixed-type IPMN, as this type has a higher association with malignancy.

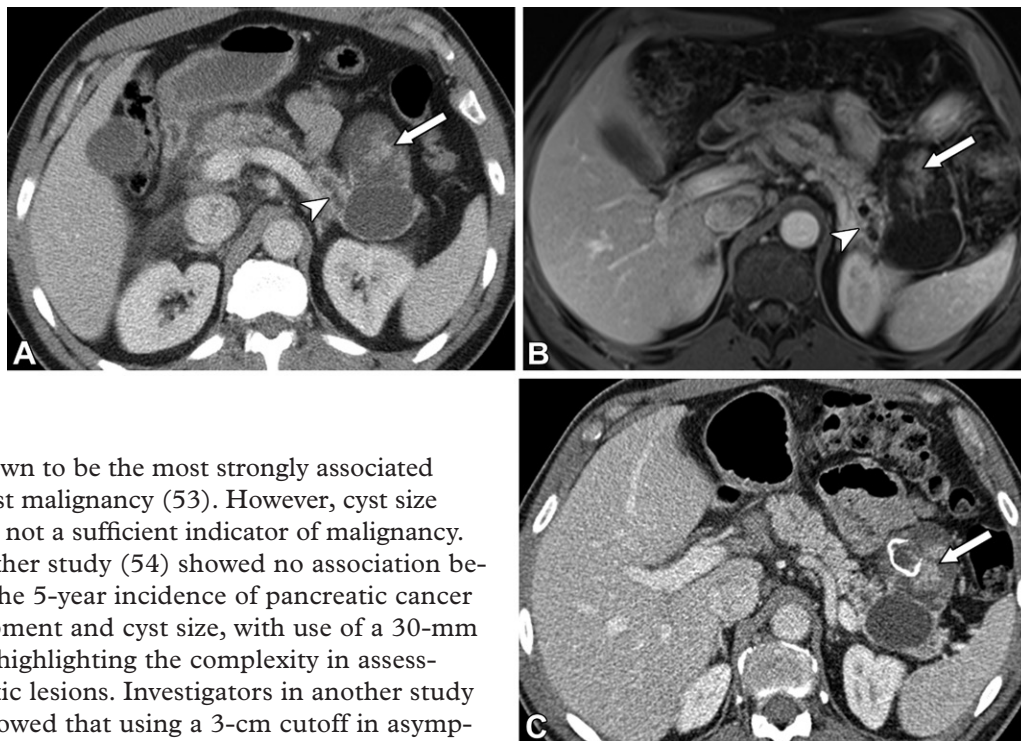
“Worrisome” Features and High-Risk Stigmata

The imaging features of IPMNs that were reported as “worrisome” for malignancy in the 2017 international consensus Fukuoka guidelines and suggest the need for further investigation with endoscopic US include a cyst diameter greater than 3 cm, an enhancing mural nodule smaller than 5 mm, a thickened enhancing cyst wall, MPD dilatation of 5–9 mm, an abrupt change in MPD diameter with upstream pancreatic atrophy, lymphadenopathy, and a cyst growth rate of greater than or equal to 5 mm per 2 years (42). The American College of Radiology guidelines suggest using an MPD diameter threshold of 7 mm (23).

Clinically, an elevated carbohydrate antigen 19-9 level also suggests malignancy (42). The presence of a solid component or a mural nodule in a cystic lesion is highly associated with malignancy in IPMNs and MCNs, irrespective of the subtype or cyst size (8). It is critical to have high suspicion for malignancy in cases of MCNs or IPMNs with a solid component, even in the setting of pancreatitis, in which a fluid collection might be suspected to be a pseudocyst or walled-off necrosis (Fig 13). High-risk stigmata considered by the International Association of Pancreatology to be indications for resection include obstructive jaundice, a 5-mm or larger enhancing mural nodule, and MPD dilatation of 1 cm or greater without another cause of obstruction (Fig 14) (42). In addition, recent studies have shown that dilatation of the dominant uncinat duct may be an independent risk factor for high-grade dysplasia or invasive carcinoma in patients with IPMN (51,52). AlMasri et al (52) showed that the risk of high-grade dysplasia or invasive carcinoma associated with dilatation of the uncinat duct persisted even with distant IPMNs in the dorsal pancreas.

As noted earlier, the size of cystic lesions is one factor that has been used as a discriminatory feature, with larger lesions (typically >3 cm) considered to be more worrisome, having greater malignant potential and requiring more aggressive management according to the guidelines. This was confirmed in a subsequent meta-analysis of cyst features, in which cyst size larger than 3 cm

Figure 13. Adenocarcinoma arising in IPMN in a 53-year-old man with pancreatitis and suspected of having walled-off necrosis. (A) Axial contrast-enhanced CT image shows mild peripancreatic inflammatory changes and a bilobar 7.4-cm complex cystic mass (arrow) with central soft-tissue attenuation arising from the pancreatic tail that was believed to represent walled-off necrosis. (B) Axial contrast-enhanced T1-weighted MR image shows the lobulated cystic mass (arrow) with internal enhancing soft tissue. In the setting of pancreatitis, this mass was initially believed to represent walled-off pancreatic necrosis. The lesion appears to communicate with a prominent pancreatic duct (arrowhead in A and B). (C) Axial contrast-enhanced CT image 8 months later again shows the complex cystic mass (arrow) with a persistent soft-tissue component and a stent from prior cystogastrostomy. The patient underwent distal pancreatectomy, splenectomy, and partial gastrectomy. Histopathologic analysis revealed multifocal grade 2 ductal-type invasive adenocarcinoma arising in an IPMN.



was shown to be the most strongly associated with cyst malignancy (53). However, cyst size alone is not a sufficient indicator of malignancy.

Another study (54) showed no association between the 5-year incidence of pancreatic cancer development and cyst size, with use of a 30-mm cutoff, highlighting the complexity in assessing cystic lesions. Investigators in another study (55) showed that using a 3-cm cutoff in asymptomatic patients would have been inaccurate for indicating premalignant or malignant disease in 14% of patients, and this cutoff led to 28% of patients with cysts larger than 3 cm undergoing unnecessary surgery (55). In addition to baseline cyst size, cyst growth rate has also been suggested as a “worrisome feature” at imaging (42). However, there is conflicting data on the ability of cyst growth, analogous to cyst size, to predict malignancy. While multiple studies have shown an association between cyst growth and malignancy, in a relatively recent study, El Chafic et al (56) did not find such an association between rapid cyst growth and advanced neoplasia at surgical-pathologic analysis (56). The uncertainty regarding the significance of large cyst size and fast growth rate in predicting malignancy highlights the potential role of cyst aspiration in the management of patients with these findings.

Field Defect

Assessing the risk of a single pancreatic cyst developing into invasive cancer can be challenging, and the presence of an IPMN may portend an increased risk of invasive pancreatic cancer at disparate sites within the pancreas (Fig 15). This

field defect suggests that an IPMN may be an indicator of background parenchymal abnormality, akin to cirrhosis leading to the development of hepatocellular carcinoma, esophageal dysplasia or Barrett esophagus leading to esophageal carcinoma, or in the breast, lobular carcinoma in situ being a marker of future increased risk for cancer. Multiple studies (57–60) have shown the increased prevalence of synchronous and metachronous invasive adenocarcinoma during the follow-up of IPMNs (Figs 16–18). In the Ingkakul et al (58) study, synchronous or metachronous concomitant invasive pancreatic carcinoma was detected in 9.3% of patients. Consequently, in addition to closely assessing the known pancreatic cysts, it is important to fully examine the remaining pancreatic tissue.

Study investigators have also looked at risk factors associated with IPMN concomitant with invasive carcinoma. Age older than 70 years (57), new-onset or worsening diabetes mellitus, and abnormal serum carbohydrate antigen 19–9 levels (58) have been shown to be associated with invasive carcinoma. Ikegawa et al (61) showed

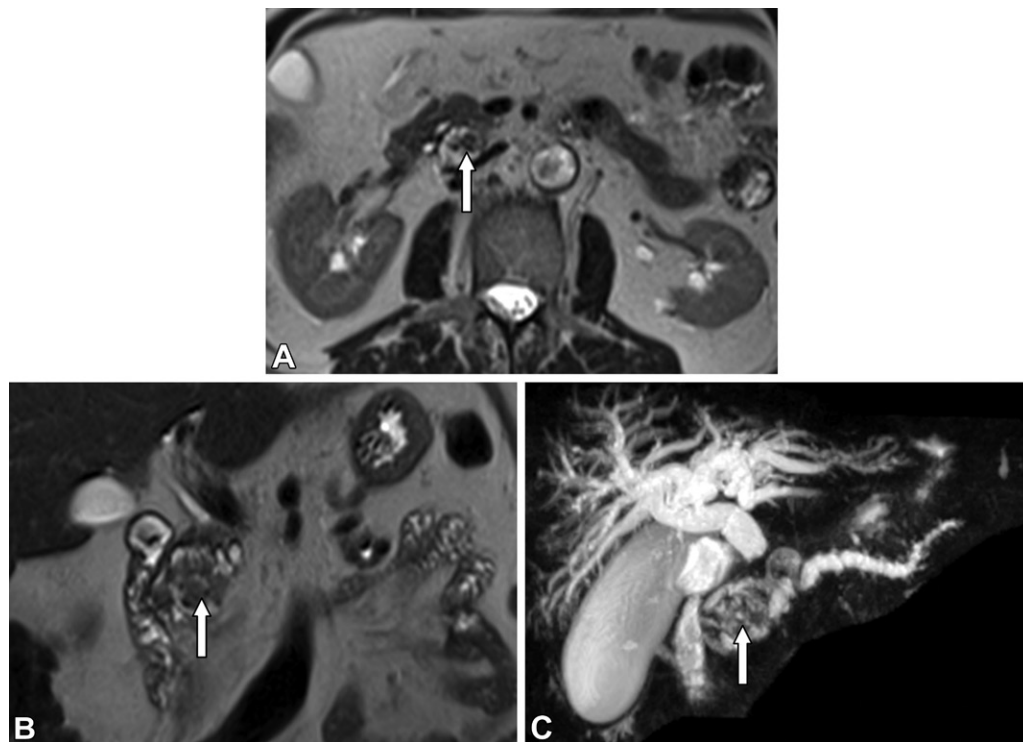


Figure 14. Adenocarcinoma in a 72-year-old man with a history of left lower quadrant pain. (A) Axial T2-weighted MR image shows a partially cystic 1.9-cm lesion arising from the uncinate process of the pancreas, with a 0.7-cm mural nodule (arrow). (B) Coronal T2-weighted MR image 4 years later shows that the solid component of the mass (arrow) has enlarged to 3.4 cm. Fine-needle aspiration biopsy yielded adenocarcinoma arising from a main-duct IPMN. The patient opted for nonsurgical management. (C) Follow-up MRCP image 2 years later shows a larger (3.6-cm) solid mass (arrow) in the dilated MPD, and there is now intrahepatic and common bile duct dilatation.

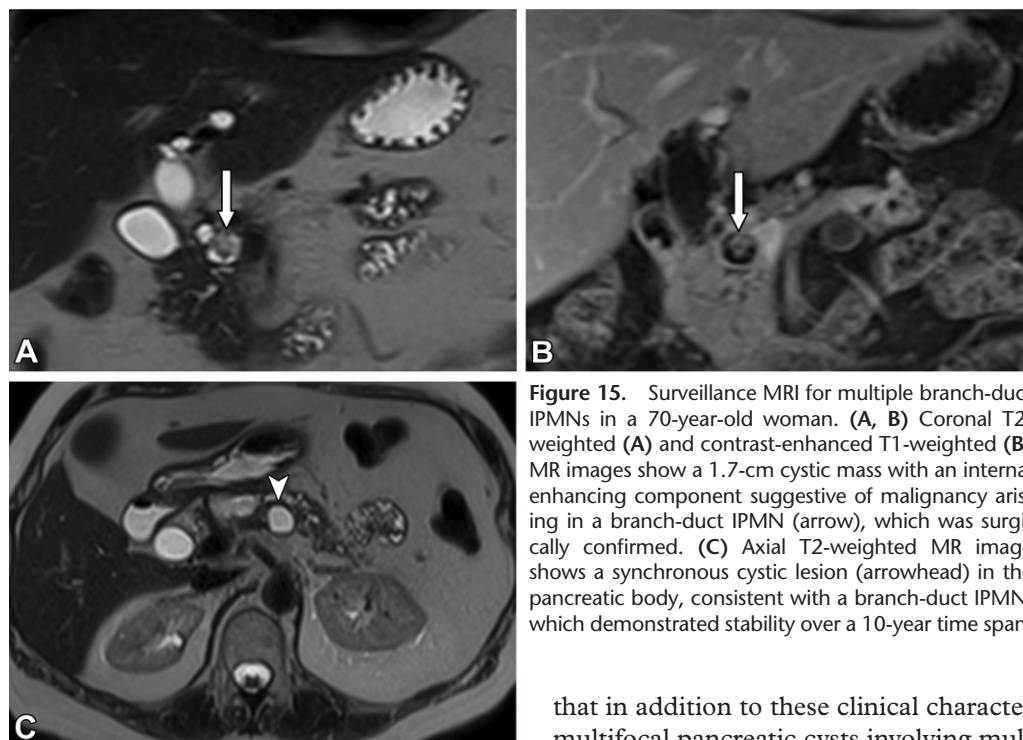


Figure 15. Surveillance MRI for multiple branch-duct IPMNs in a 70-year-old woman. (A, B) Coronal T2-weighted (A) and contrast-enhanced T1-weighted (B) MR images show a 1.7-cm cystic mass with an internal enhancing component suggestive of malignancy arising in a branch-duct IPMN (arrow), which was surgically confirmed. (C) Axial T2-weighted MR image shows a synchronous cystic lesion (arrowhead) in the pancreatic body, consistent with a branch-duct IPMN, which demonstrated stability over a 10-year time span.

that in addition to these clinical characteristics, multifocal pancreatic cysts involving multiple regions of the pancreas also were a risk factor for IPMN and invasive carcinoma.

This field defect further complicates the already complex follow-up guidelines for IPMNs.

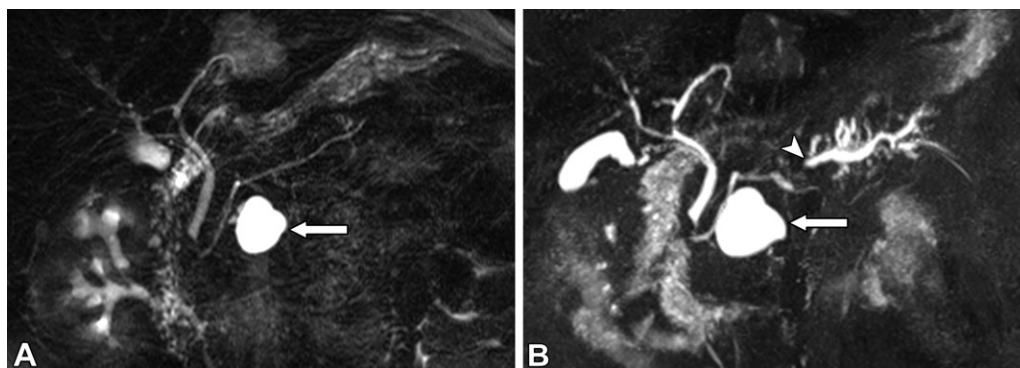


Figure 16. Surveillance of branch-duct IPMN in a 79-year-old man. (A) MRCP image shows a cystic 2.2-cm lesion (arrow) in the pancreatic head without internal septa or nodularity, which was found to be branch-duct IPMN. The patient elected not to undergo additional surveillance, given his advanced age. However, he underwent MRI for abdominal discomfort 1 year later. (B) MRCP image shows a slight increase in the size (2.4 cm) of the pancreatic head cystic lesion (arrow) and new dilatation of the MPD, with abrupt ductal termination at the level of the pancreatic body (arrowhead) being suspicious for tumor. The patient underwent endoscopic US and biopsy, followed by distal pancreatectomy and splenectomy. Surgical-pathologic analysis yielded poorly differentiated adenocarcinoma with negative margins and no lymphovascular invasion. The patient is alive 8 years later. This case illustrates the field defect associated with IPMNs and associated pancreatic carcinoma.



Figure 17. Pancreatic adenocarcinoma in an 81-year-old man with a history of jaundice, and clinical findings of gastric outlet obstruction. (A, B) Coronal contrast-enhanced CT (A) and T2-weighted MR (B) images show a mass (arrow) in the pancreatic head, with dilatation of the common duct secondary to the mass. (C) MRCP image shows branch-duct IPMNs in the pancreatic tail (arrowhead) and head (arrow). Pathologic analysis yielded undifferentiated pancreatic ductal adenocarcinoma and scattered areas of low-grade IPMN. The patient subsequently underwent a Whipple procedure. This case highlights the potential for simultaneous poorly differentiated adenocarcinoma and low-grade IPMNs.

It has been shown that patients with IPMNs have, in addition to an increased risk of long-term invasive carcinoma, an increased risk of invasive carcinoma even after resection of the IPMN, including IPMNs with low-grade dysplasia (62–64). He et al (62) found that a family

history of pancreatic cancer was the only factor associated with recurrence following IPMN resection, potentially highlighting a subgroup that requires more attention at follow-up. However, concern that even annual surveillance might not be sufficient to detect early, resectable invasive

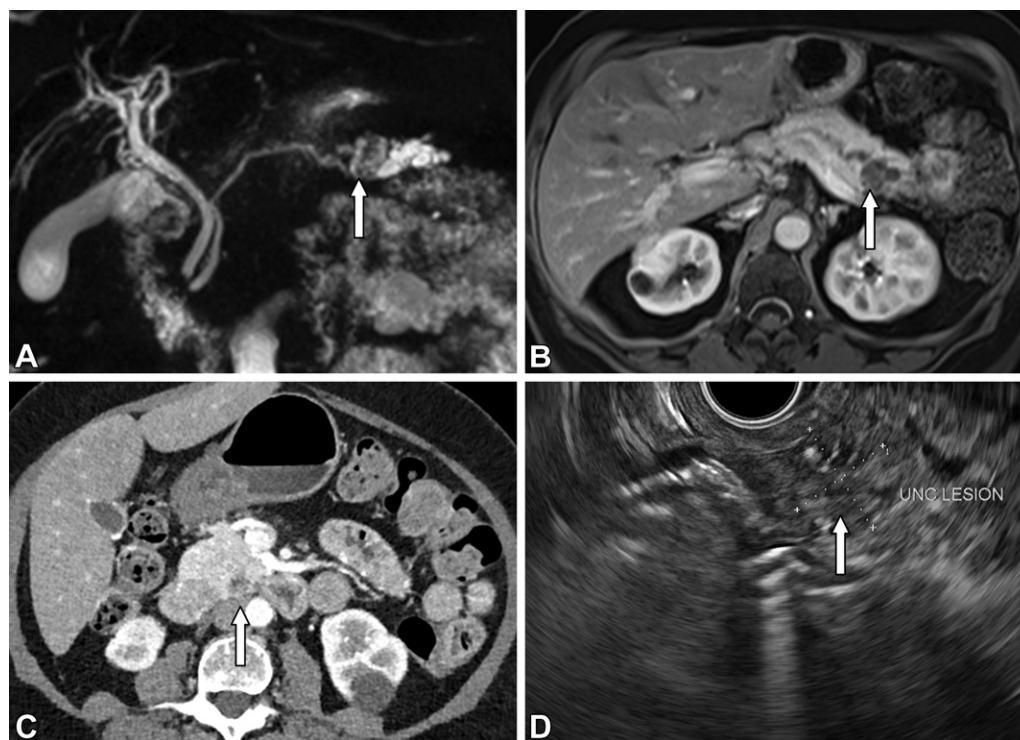


Figure 18. Pancreatic adenocarcinoma in a 70-year-old woman with pancreatitis and a pancreatic tail lesion seen on CT scans, illustrating a field defect and metachronous lesion. (A) MRCP image shows a cystic lesion in the tail involving the MPD with a filling defect (arrow). (B) Axial contrast-enhanced T1-weighted MR image shows a dilated pancreatic duct with a hypoenhancing 2.7-cm mass (arrow). Distal pancreatectomy yielded minimally invasive adenocarcinoma arising in the background of the main-duct IPMN with extensive high-grade dysplasia and carcinoma in situ. (C) Axial contrast-enhanced CT image 1 year later shows a heterogeneous 1.7-cm hypoenhancing lesion (arrow) in the uncinete process. (D) Endoscopic US image shows a solid lesion (arrow). The patient underwent a Whipple procedure, which yielded multiple foci of invasive adenocarcinoma, which was well to moderately differentiated with mucinous features, arising in the background of the IPMN.

carcinoma is most problematic (Fig 19). Ohtuska et al (64) found patients with unresectable invasive carcinoma more than 10 years after IPMN resection and recommended close long-term follow-up after resection.

Follow-up Guidelines

Given the high prevalence of pancreatic cystic lesions, which are increasingly being detected with high-spatial-resolution cross-sectional imaging, appropriate and systematic follow-up guidelines are necessary. Multiple societies have created guidelines for the management of pancreatic cysts. Five major guidelines are the international consensus Fukuoka 2017, American College of Gastroenterology 2018, American College of Radiology 2017, European Study Group 2018, and American Gastroenterology Association 2015 guidelines. However, as shown in Table E1 (13,23,42,65,66), there is variability in the guidelines among the different societies.

From a clinical perspective, the guidelines vary with respect to the target patient population, incorporation of clinical symptoms, and timing and duration of the initial and long-term imaging

surveillance (45,67). Also, from an imaging perspective, these guidelines highlight slightly different features associated with malignancy (45,67). The differing approach to long-term surveillance is especially notable. In both the 2015 American Gastroenterology Association guidelines and 2017 American College of Radiology guidelines, there is a set end point for surveillance for cases of cyst stability, at 5 and 10 years, respectively. The other guidelines do not recommend a set end point for surveillance.

However, multiple studies (46,68–70) have shown that patients with IPMNs have a long-term risk of developing invasive cancer, extending beyond the 5-year timeline suggested by the American Gastroenterology Association. Considering this long-term risk, the optimal duration of follow-up is unclear. Multiple authors, including Oyama et al (46), who suggest that these patients should be followed for 20 years or longer, now suggest that follow-up should exceed 5 years. Because of the large number of people with incidental pancreatic cysts, many of which are indolent, the associated potential costs of surveillance will be considerable.

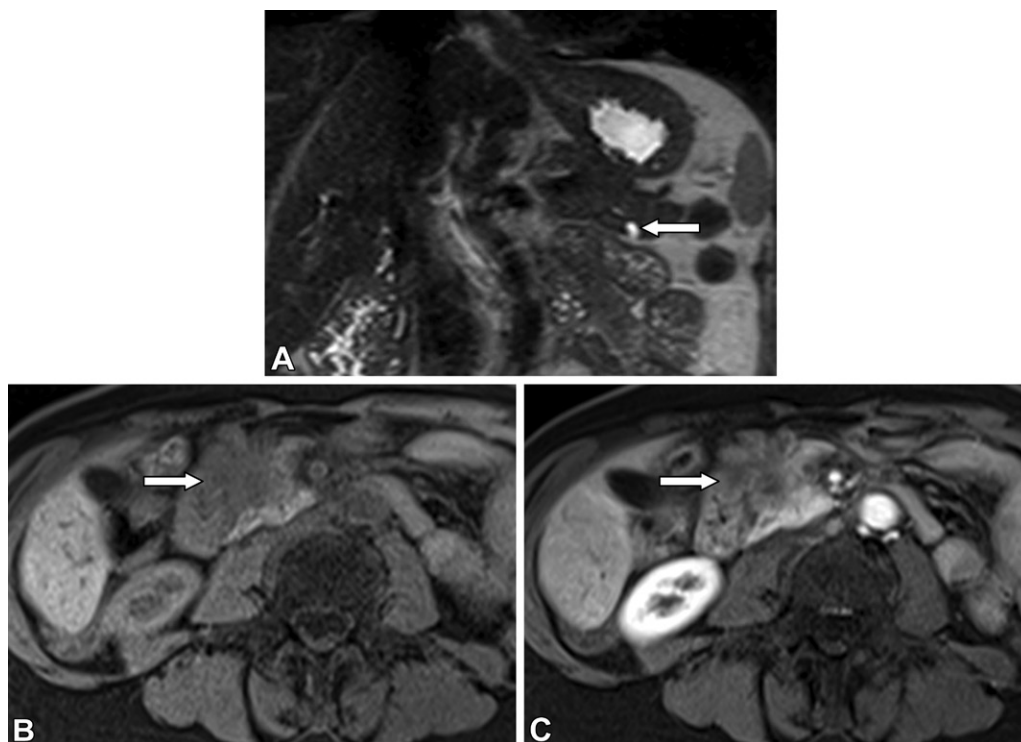


Figure 19. Surveillance for branch-duct IPMN in a 61-year-old woman. (A) Coronal T2-weighted MR image shows a small T2-hyperintense cystic lesion (arrow) in the pancreatic tail that had decreased in size compared with the prior study (not shown). It had been previously noted to communicate with the pancreatic duct, suggesting a branch-duct IPMN. (B, C) Axial noncontrast T1-weighted (B) and contrast-enhanced (C) MR images 1 year later show a heterogeneously enhancing mass (arrow) in the pancreatic head. Fine-needle aspiration biopsy yielded adenocarcinoma with mucinous features. This case illustrates the field defect and the challenges with surveillance for even tiny suspected branch-duct IPMNs.

As a result of the differences in guidelines, key stakeholders (eg, radiologists, gastroenterologists, and surgeons) must review and come to a consensus regarding which guideline (or combination of guidelines) to follow at their individual institutions. We developed our institutional pancreatic cyst guidelines after reviewing the available medical literature by using modified Delphi methodology and reaching a consensus on the proposed recommendations. The guidelines of each society were provided in a table based on cyst size, and our panel members were asked to “agree” or “disagree” with each statement and provide comments. This method was used until an agreement was reached, with the final step consisting of a face-to-face meeting with the main contributors.

Subsequently, the proposed guidelines were sent to all gastroenterology, surgery, and radiology clinicians with an interest in pancreatic imaging for comments, with subsequent revisions performed until agreement was reached. Given the differences in the guidelines reported in the literature, our final institutional guidelines represent a synthesis of the available literature and guidelines, with recommended follow-up intervals and durations. On the basis of these

institutional consensus guidelines, we created a “macro” to standardize our follow-up recommendations according to cyst size, similar to the approach used to develop the societal guidelines. We used this macro standardization to minimize reporting inconsistencies among radiologists, which are an issue that has been reported to be problematic (71,72).

Abbreviated MRI Protocols

With increasing detection of small pancreatic cystic lesions and potential lifelong follow-up for patients with IPMNs, a short limited protocol and noncontrast MRI are enticing. Studies have shown no benefit to performing cyst assessment with the addition of gadolinium-based contrast material during surveillance imaging (73–75), and noncontrast MRI may be preferable to patients, considering increasing concerns about long-term gadolinium retention. Pozzi-Mucelli et al (75) found that short-protocol MRI with use of only axial and coronal T2-weighted half-Fourier single-shot turbo spin-echo sequences, as well as an axial noncontrast T1-weighted three-dimensional gradient-echo fat-suppressed sequence, demonstrated performances equivalent to that of contrast-enhanced MRI in cyst follow-up.

The decreased cost and time achieved with such a short-protocol MRI examination, as compared with standard-protocol MRI, is also advantageous owing to the increased detection and reporting of pancreatic cysts and the substantial number of follow-up MRI examinations required. However, while gadolinium-enhanced MRI examinations, as compared with short-protocol MRI studies, have been evaluated for the follow-up of known pancreatic cystic lesions, it remains unclear whether these short noncontrast protocols affect the identification of synchronous or metachronous invasive carcinoma, which may be subtle without gadolinium-based contrast medium administration and is of concern given the potential for field defect.

The limitations of using a short MRI protocol include decreased sensitivity in the detection of solid lesions in patients whose MR images show a low-T1-signal-intensity pancreas associated with atrophy and chronic pancreatitis. Given the increasing understanding of the field defect and the risk of carcinoma in the background pancreas, further studies are needed. Until there is more evidence or guidance, our institution continues to perform contrast-enhanced MRI examinations.

Conclusion

Incidental pancreatic cystic lesions are being detected more frequently on imaging studies. An organized approach to interpreting imaging studies to determine the diagnosis is critical. Those cysts that cannot be diagnosed may require follow-up imaging or aspiration. Expert consensus guidelines from different societies have been developed to promote standardized management and follow-up. However, these guidelines are inconsistent and evolving according to additionally published scientific literature. Institutional standardized reporting and macros are helpful for ensuring the uniformity of interpretations.

The identification of worrisome features and/or high-risk stigmata is important, as these findings may convey the need for aspiration or surgery. Newer concepts such as field defect also are important considerations that may affect management. Radiologists play a critical role in the detection and characterization of pancreatic cystic lesions, in the follow-up recommendations for these lesions, and in the detection of cancer.

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